



THE COPPERBELT UNIVERSITY

**SCHOOL OF INFORMATION COMMUNICATION TECHNOLOGY
(SICT)**

CS301 – PROJECT MANAGEMENT

TITLE:

PROJECT PROPOSAL (TAX-OPOLGY: THE PUBLIC GOOD GAME)

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1.0 BACKGROUND

In an era where economic literacy is crucial for informed citizenship, traditional educational tools often fall short in engaging learners with complex concepts such as taxation and public goods. Taxes are frequently perceived as burdensome obligations rather than investments in societal well-being, leading to widespread misconceptions about their role in fostering economic growth and community development. This gap in understanding is particularly evident in developing economies like Zambia, where public discourse on fiscal policy can influence civic participation and compliance.

Existing games, such as classic Monopoly, emphasize individual wealth accumulation and competition, often at the expense of collective benefits, reinforcing a zero-sum mindset that does not reflect the interdependent nature of modern societies. However, Monopoly and similar board games have long been recognized as effective tools for teaching foundational concepts in economics, negotiation, and strategy. Their interactive, rule-based environments enable experiential learning—players absorb principles of resource management, risk assessment, and market dynamics through play. This demonstrates the potential of game-based learning not only to educate but also to shape perceptions and behaviors, offering a pathway to redesign gameplay for prosocial outcomes such as cooperation and civic responsibility.

The concept of Tax-opoly emerges as a response to this challenge, reimagining the Monopoly framework to model tax compliance as a strategic choice with tangible communal outcomes. Originally conceptualized as a physical board game, Tax-opoly shifts the focus from bankrupting opponents to collaboratively building a high Quality of Life Index (QLI) through tax adherence and public investments. By simulating real-world dynamics like the risks of evasion via audits and the multipliers of upgraded infrastructure, the game educates players on the “free rider” problem, the value of transparency, and the virtuous cycle of taxation leading to shared prosperity. With the rise of digital platforms, transforming Tax-opoly into an interactive online game presents an opportunity to scale its impact, making it accessible to a global audience, including students, policymakers, and the general public, while incorporating modern features like multiplayer modes and data analytics for enhanced learning.

1.1 INTRODUCTION

Revenue generation through taxation is essential for national development, providing the financial foundation for public services such as education, healthcare, infrastructure, and security. For countries like Zambia, improving tax compliance is not only a fiscal imperative but also a cornerstone of sustainable growth and social equity. However, the effectiveness of taxation depends heavily on public understanding and voluntary participation—when citizens perceive taxes as fair, transparent, and beneficial, compliance improves, strengthening the social contract between the state and its people.

Educating the public on the importance of tax compliance is therefore critical, yet conventional approaches often fall short. Traditional methods—such as lectures, pamphlets, or static media—tend to be abstract, passive, and poorly suited to conveying the tangible links between tax contributions and community well-being. These methods frequently fail to engage learners, especially younger and digitally-oriented audiences, and do little to counteract perceptions of taxation as a punitive or opaque burden. As a result, misinformation and skepticism persist, hindering efforts to foster a culture of fiscal responsibility and shared investment.

In response to these challenges, this project proposes the development of "**Tax-opoly: The Public Good Game**"—a digital playable application designed to transform how tax literacy is taught and understood. Building on proven educational game design, Tax-opoly creates an immersive, interactive experience that turns abstract economic principles into engaging gameplay. Players balance personal savings with contributions to a shared Public Treasury, which funds upgrades to essential services like schools, hospitals, and transportation systems. Through a blend of strategy, risk, and cooperation, participants learn that true success lies in collective advancement rather than isolated gains.

The digital version will leverage technology to enhance accessibility and engagement, featuring cross-platform compatibility, real-time multiplayer options, customizable scenarios for workshops, and integrated analytics to track learning outcomes. By proposing this project, we seek to bridge the divide between economic theory and practical understanding, ultimately promoting a culture of informed fiscal responsibility and societal investment.

2.0 PROBLEM STATEMENT

In many societies, particularly in developing nations like Zambia, tax compliance remains a persistent challenge due to widespread perceptions of taxation as an opaque, punitive mechanism rather than a vital tool for collective progress. According to recent reports from the Zambia Revenue Authority, tax evasion and informal economic activities contribute to significant revenue shortfalls, hindering investments in essential public services such as education, healthcare, and infrastructure. This issue is exacerbated by low levels of financial literacy among the general population, where traditional educational approaches; lectures, pamphlets, and static media, fail to engage learners or demonstrate the direct link between tax payments and societal benefits. As a result, citizens often view taxes through a lens of individualism, overlooking the "free rider" problem and the multiplier effects of public goods on economic growth.

The absence of interactive, experiential learning tools perpetuates this disconnect, especially in a digital age where younger generations and remote communities seek accessible, gamified content. Classic games like Monopoly, while popular, reinforce competitive, zero-sum dynamics that do not model the cooperative aspects of fiscal policy. This gap leaves a critical need for an innovative solution that can educate diverse audiences on the value of tax adherence, fostering trust in government systems and encouraging voluntary compliance. By addressing these shortcomings, Tax-opoly as a digital game has the potential to transform public understanding, but its current conceptual form limits scalability and reach in resource-constrained environments.

3.0 OBJECTIVES

3.1 General Objective

To develop and evaluate a functional digital prototype of *Tax-opoly: The Public Good Game* as a university project, demonstrating the feasibility of transforming the original board game concept into an interactive educational tool for improving understanding of tax compliance and public goods.

3.2 Specific Objectives

1. To gather system and user requirements for the digital Tax-opoly prototype using methods such as questionnaires, observation, and informal interviews with selected university students.
2. To analyse the collected requirements in order to identify key functional, educational, and usability requirements for the system.
3. To design the digital Tax-opoly system using standard system design techniques, including use-case diagrams, game flow diagrams, and wireframes, to model gameplay and user interaction at a conceptual level.
4. To develop a functional digital prototype using appropriate, cost-effective, and open-source development tools suitable for a student-led project.
5. To test and evaluate the prototype through manual testing and playtesting sessions involving 20–50 university peers, focusing on functionality, usability, and educational effectiveness.
6. To assess the feasibility and learning impact of the prototype using qualitative feedback and simple survey-based evaluation methods.
7. To document the development process, testing outcomes, challenges encountered, and recommendations for future improvements in a comprehensive project report.

4.0 SCOPE

This project focuses on demonstrating the feasibility of a digital Tax-opoly prototype as an educational tool, with a deliberately limited scope to align with the constraints of a student-led initiative, including time, resources, and technical expertise. The project will prioritize core functionality to prove the concept while excluding advanced features that would require extensive development or external dependencies.

4.1 In-Scope

The following items are considered within the scope of this project:

- Identification and specification of high-level functional requirements necessary to support core gameplay and educational objectives.
- Conceptual modelling of gameplay elements related to taxation decisions, public treasury contributions, and quality-of-life outcomes, without detailing internal algorithms or implementation logic.
- Integration of basic educational components, including tutorials, in-game prompts, and end-of-game debriefs, to explain concepts such as tax compliance, public goods, and collective investment.
- Development of a functional digital prototype suitable for demonstration and evaluation purposes.
- Manual functional testing, usability testing, and playtesting conducted with selected university peers to assess feasibility and educational effectiveness.
- Preparation of academic documentation, including requirements summaries, design artifacts, testing results, and a final project report.
- Completion of the project within one academic year by a small student team using minimal financial resources.

4.2 Out-of-Scope

The following items are explicitly excluded from the scope of this project:

- Development of a full-scale, production-ready system or commercial application.
- Advanced multiplayer features such as real-time online gameplay, server hosting, or cross-device synchronization.
- Detailed economic simulations or the use of real-world taxation data beyond simplified representations.

5.0 LITERATURE REVIEW

Financial literacy and tax compliance remain persistent challenges in many developing economies, including Zambia. Studies indicate that limited understanding of taxation systems contributes to tax evasion, mistrust in public institutions, and reduced public revenue, ultimately affecting the provision of essential public services such as education, healthcare, and infrastructure (ActionAid Zambia, n.d.). Traditional approaches to teaching taxation, including lectures and printed materials, often fail to engage learners or clearly demonstrate the relationship between tax contributions and public benefits.

Recent literature highlights gamification as an effective method for improving engagement and comprehension in financial education. Gamification applies game design elements such as challenges, feedback, and rewards to learning contexts, making abstract concepts more accessible and engaging. The Commonwealth (2017) notes that gamified financial tools increase learner participation and retention by transforming complex financial ideas into interactive experiences, particularly for younger audiences.

Experimental economic simulations, especially public goods games, have been widely used to illustrate collective action problems such as the free-rider problem and cooperation dilemmas. According to the CORE Economics Team, public goods games enable participants to experience how individual contribution decisions influence collective outcomes, making them effective tools for teaching fiscal responsibility and cooperative behaviour.

Digital games further enhance learning by offering accessibility, interactivity, and opportunities for iterative improvement. Siregar and Suryanto (2024) found that gamified learning environments significantly improve student engagement and understanding in financial education contexts. However, many existing games, such as traditional Monopoly, emphasize individual wealth accumulation and competitive success, reinforcing zero-sum thinking that does not accurately reflect real-world taxation systems.

The proposed “Tax-opoly: The Public Good Game” builds on existing research by combining gamification principles with public goods theory in a digital format. By shifting the focus from individual accumulation to collective quality-of-life outcomes, the project addresses a gap in accessible, context-aware educational tools on taxation. This study therefore demonstrates the feasibility of a low-cost, student-developed digital prototype for improving understanding of tax compliance and public investment within a university and developing-country context.

6.0 METHODOLOGY

This project will adopt a hybrid Agile–iterative development methodology, tailored to the constraints of a student-led university project and focused on feasibility, simplicity, and proof-of-concept delivery. While the project will follow a high-level sequential structure for planning and reporting purposes, development activities will be conducted using Agile principles that emphasize incremental development, continuous feedback, and adaptability to change.

The project will be executed over an academic period of approximately six to seven months and organized into clearly defined phases. Within each phase, development will proceed through short iterations (sprints), during which functional components of the system will be designed, implemented, tested, and reviewed. Feedback from playtesting and peer evaluations will be incorporated into subsequent iterations to progressively refine gameplay mechanics, usability, and educational effectiveness.

Requirements Gathering and Analysis

System and user requirements will be gathered using lightweight techniques suitable for a university environment, including questionnaires, informal interviews, and observation of potential users. The collected data will be analyzed qualitatively to identify key functional, educational, and usability requirements that will guide system design and development. Requirements will be prioritised to ensure focus on core features necessary to demonstrate feasibility.

System Design

The system will be designed at a conceptual level using standard system design techniques such as use-case diagrams, game flow diagrams, and wireframes. These artifacts will be used to model user interactions, gameplay flow, and educational elements without detailing internal algorithms or implementation specifics. Design decisions will aim to balance simplicity, clarity, and alignment with the project’s educational objectives.

Development Approach

Development will follow an incremental and iterative process aligned with Agile principles. Core gameplay functionality will be implemented first, followed by the gradual addition of supporting features in subsequent iterations. Version control tools i.e. Git and GitHub will be used to manage code changes and maintain project organization. Emphasis will be placed on delivering a minimum viable prototype that prioritizes functionality and educational value over aesthetic refinement.

Testing and Evaluation

Testing will be conducted throughout the development process to ensure system stability and usability. This will include manual functional testing and usability testing during development, followed by structured playtesting sessions involving 20 - 50 university peers. Feedback will be collected using simple survey instruments to evaluate usability, engagement, and perceived

learning outcomes. The results will be analyzed using basic descriptive methods to assess the feasibility and educational effectiveness of the prototype.

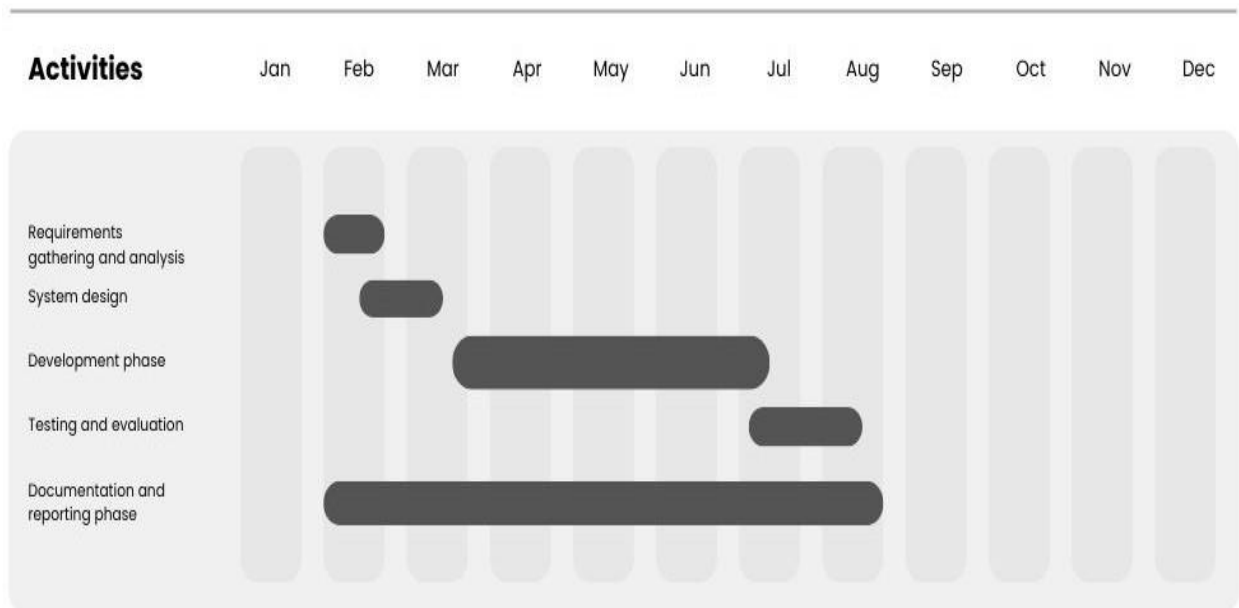
Documentation and Reporting

Comprehensive documentation will be maintained throughout the project lifecycle as required by the SICT department expectations. This will include requirements summaries, design artifacts, development logs, testing results, and evaluation findings. The project will conclude with the preparation of a final academic report that documents the development process, challenges encountered, outcomes achieved, and recommendations for future improvements, in accordance with CS301 requirements.

6.1 PROJECT TIMELINE (GANTT CHART)

The Gantt below visually depicts the project's development phase and the timelines for each of these phases to ensure requirements and expectations are met as well as submissions required for the project:

Project Timeline Gantt Chart



7.0 EXPECTED BUDGET

We intend to build “Tax-opoly: The Good Game” with mostly open source and free third-party tools consequently not requiring subscriptions for SaaS services. Moreover, other indirect costs such as internet access, electricity, and personal computing resources are assumed to be available to the team and are therefore not explicitly costed:

- | | |
|-------------------------------|------|
| 1. Printing and binding costs | K200 |
| 2. Miscellaneous costs | K100 |

8.0 TEAM RESPONSIBILITIES

- | | |
|----------------------|--------------------------------|
| 1. Lwanga Mwaba | Documentation manager |
| 2. Benjamin Chaambwa | Team Lead |
| 3. Darius Chungu | Quality Assurance (QA) Manager |
| 4. Isaac Tembo | Chief programmer |

9.0 CONCLUSION

This university project proposes the development of a digital prototype for “Tax-opoly: The Public Good Game”, transforming an innovative board game concept into an accessible, interactive tool to enhance understanding of tax compliance and public goods. By addressing the gap in engaging financial literacy education through gamification, the prototype will demonstrate how strategic gameplay can illustrate complex economic principles; such as the risks of tax evasion, the free-rider problem, and the multiplier effects of collective investments, in a relatable and enjoyable format.

Within the constrained scope of a student-led initiative, the project is designed to be feasible, utilizing free tools and a structured 6-month timeline to deliver a functional, testable prototype. Through iterative design, development, playtesting with peers, and thorough documentation, it will provide tangible evidence of the concept's viability while offering valuable insights into digital game development for educational purposes.

Ultimately, this prototype not only fulfills academic requirements but also lays the foundation for a potentially impactful tool that could promote fiscal awareness among students and broader audiences. Successful completion will validate the power of gamified learning in shifting perspectives on taxation from obligation to shared investment, paving the way for future expansions and real-world applications in financial education.

10.0 REFERENCES

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